

titleLocality Tradeoffs for 2D Quantum LDPC Codes (a BPT Restatement with Explicit Parameters)

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Abstract

We study tradeoffs between rate, distance, and geometric locality for 2D quantum LDPC stabilizer codes. Our main theorem is a restatement of the 2D Bravyi–Poulin–Terhal (BPT) bound that makes the dependence on interaction range and maximum stabilizer weight explicit. We also contrast these limitations with recent asymptotically good (necessarily non-geometrically-local) quantum LDPC constructions, and we explain the role of separator phenomena in the locality barrier.

1 Introduction

Quantum error correction is fundamental to fault-tolerant quantum computation. Among the various code families, quantum low-density parity-check (LDPC) codes have emerged as particularly promising candidates due to their favorable scaling properties and potential for efficient decoding.

The central question in quantum LDPC code theory concerns the tradeoffs between code parameters:

- Rate $R = k/n$ (logical qubits per physical qubit)
- Distance d (number of errors the code can correct)
- Maximum check weight w (sparsity of parity checks)
- Geometric locality constraints

Classical results such as the Bravyi–Poulin–Terhal (BPT) bound [BPT10] establish fundamental limits for geometrically local codes. This note focuses on the 2D setting and keeps track of how the relevant constants scale with an explicit interaction range and stabilizer weight.

1.1 Main Contributions

In this paper, we:

1. Rephrase the 2D Bravyi–Poulin–Terhal (BPT) locality tradeoff in terms of rate, interaction range, and maximum stabilizer weight
2. Highlight the role of expansion in non-local constructions and why it is incompatible with fixed-dimensional geometric locality
3. Demonstrate tightness in 2D via the toric code

2 Preliminaries

2.1 Quantum LDPC Codes

Definition 2.1 (CSS Code Construction). A Calderbank-Shor-Steane (CSS) code is a particular type of quantum error-correcting code constructed from two classical linear block codes, C_1 and C_2 , with parity check matrices H_X and H_Z , such that $C_2^\perp \subseteq C_1$. This condition is equivalent to the orthogonality constraint $H_X H_Z^T = 0$. The resulting quantum code corrects X -errors using H_Z and Z -errors using H_X .

Definition 2.2 (Quantum LDPC Code). A quantum LDPC code is a CSS code $\mathcal{C} = [[n, k, d]]$ defined by sparse parity-check matrices H_X and H_Z satisfying $H_X H_Z^T = 0$, with bounded stabilizer weight and bounded qubit degree. Concretely, suppose each row of H_X and H_Z has Hamming weight at most $w = O(1)$ (maximum stabilizer/check weight) and each column has Hamming weight at most $q = O(1)$ (maximum number of checks acting on a qubit). Both bounds are independent of the block length n .

Definition 2.3 (Tanner Graph). The Tanner graph $G = (V \cup C, E)$ of a quantum LDPC code has variable nodes V corresponding to qubits and check nodes C corresponding to stabilizer generators, with edges indicating non-trivial support.

2.2 Geometric Locality

Definition 2.4 (Geometric Locality). A quantum code is *geometrically local* in $\mathbb{R}^D$ with interaction range L and density ρ if:

1. There exists an embedding of the n qubits into $\mathbb{R}^D$ such that every ball of unit volume contains at most ρ qubits.
2. Each stabilizer generator acts only on qubits contained within a ball of diameter L .

For asymptotic analysis, we typically assume $\rho = O(1)$ and define locality by the constraint that L is constant independent of n .

Remark 2.5 (Bookkeeping for L and w). In a bounded-density embedding, a ball of diameter L contains at most $O(\rho L^D)$ qubits. Thus, in many geometric models one has an implicit bound $w \leq O(\rho L^D)$ for strictly local stabilizer generators. In this note we keep w and L as explicit parameters to cleanly separate “range” from “weight,” but one should keep in mind that they may not be independent in a given hardware model.

2.3 Expander Graphs and Spectral Methods

Definition 2.6 (Spectral Expansion). A graph G is a λ -expander if the second largest eigenvalue of its normalized adjacency matrix satisfies $|\lambda_2| \leq \lambda$.

Lemma 2.7 (Expander Mixing Lemma). For a λ -expander G with n vertices and (regular) degree r , and any two vertex sets $S, T \subseteq V$:

$$\left| |E(S, T)| - \frac{r|S||T|}{n} \right| \leq \lambda r \sqrt{|S||T|}$$

3 Main Results

3.1 Tradeoff Bounds Under Combined Constraints

Theorem 3.1 (Main Tradeoff Bound (2D, BPT)). Consider a 2D geometrically local stabilizer code on n qubits with interaction range L and maximum stabilizer weight w . Let $R = k/n$ be

the rate and d the distance. Then

$$d\sqrt{R} \leq C w L,$$

for a universal constant C .

Proof. This is a restatement of the 2D BPT tradeoff for local commuting-projector / stabilizer codes [BPT10], with explicit dependence on interaction range L and maximum stabilizer weight w .

One convenient way to track the range parameter is via a coarse-graining step: partition the plane into squares of side length $\Theta(L)$, treating each square as a “supersite.” Under bounded density, each supersite contains $O(L^2)$ qubits. A stabilizer supported in a ball of diameter L intersects only $O(1)$ supersites, so the model reduces to a constant-range local code on $\Theta(n/L^2)$ supersites, at the cost of increasing the local Hilbert space dimension (and potentially the effective interaction strength/weight constants) by a factor polynomial in L . In this bookkeeping, one obtains a bound of the form

$$kd^2 \leq O((wL)^2 n),$$

and the claim follows by dividing by n and substituting $R = k/n$. In the form used here, BPT gives

$$kd^2 \leq O((wL)^2 n).$$

Dividing by n and using $R = k/n$ yields $Rd^2 \leq O((wL)^2)$, i.e., $d\sqrt{R} \leq CwL$. $\square$

Corollary 3.2. *For 2D geometrically local codes ($D = 2$) with bounded check weight w and interaction range L :*

$$d \cdot \sqrt{k} \leq O(wL\sqrt{n}) \implies kd^2 \leq O(w^2 L^2 n)$$

recovering the standard BPT bound [BPT10] with explicit dependence on w and L .

3.2 Expansion and Linear Distance (Classical)

Theorem 3.3 (Expander-Code Distance from Unique-Nighbor Expansion). *Let $C \subseteq \{0, 1\}^n$ be a classical linear code with parity-check matrix H whose Tanner graph is (d_v, d_c) -regular (variable degree d_v , check degree d_c). Suppose there exists a constant $\alpha > 0$ such that every set S of variable nodes with $|S| \leq \alpha n$ satisfies*

$$|N(S)| > \frac{d_v}{2}|S|.$$

Then the (classical) minimum distance of C is greater than αn .

Proof. Let $x \in C$ be a nonzero codeword with support $S \subseteq [n]$. Since $Hx = 0$, every check node in $N(S)$ is incident to an even number of variables from S . In particular, there can be no check node in $N(S)$ with exactly one neighbor in S .

Assume for contradiction that $|S| \leq \alpha n$. If every check in $N(S)$ has at least two neighbors in S , then counting edges between S and $N(S)$ gives

$$d_v|S| = |E(S, N(S))| \geq 2|N(S)|,$$

so $|N(S)| \leq (d_v/2)|S|$, contradicting the hypothesis. Therefore no nonzero codeword can have weight $\leq \alpha n$, and the classical distance satisfies $d > \alpha n$. $\square$

Remark 3.4. For a CSS code specified by (H_X, H_Z) , expansion conditions like Theorem 3.3 can lower bound the distances of the *underlying classical* codes defined by H_X and H_Z . Translating this into a lower bound on the *quantum* distance additionally requires controlling the quotient by the stabilizer group (e.g., excluding low-weight vectors in $\ker(H_X)$ that lie outside $\text{rowspan}(H_Z)$).

4 Tightness and Examples

4.1 Toric Codes

Example 4.1 (Tight Construction for Local Codes). Consider the standard Toric code defined on an $\ell \times \ell$ lattice on a torus. The code parameters are:

- Physical qubits $n = 2\ell^2$
- Logical qubits $k = 2$
- Distance $d = \ell = \sqrt{n/2}$
- Rate $R = k/n = 1/\ell^2$

This construction is strictly geometrically local in 2D with interaction range $L = O(1)$ and check weight $w = 4$. Applying our Main Tradeoff Bound (Theorem 3.1):

$$d \cdot R^{1/2} = \ell \cdot \left(\frac{1}{\ell^2}\right)^{1/2} = \ell \cdot \frac{1}{\ell} = 1$$

Since the right-hand side of Theorem 3.1 is $C \cdot w \cdot L = O(1)$ for a 2D local code with constant interaction range, the Toric code saturates the bound up to constants. This confirms the tightness of our result for local codes. Note that codes beating this bound (such as expander-based hypergraph product codes) must violate the geometric locality constraint.

4.2 Breaking the Barrier: Non-Local Codes

Recent breakthroughs by Panteleev and Kalachev [PK22] and others have produced "good" quantum LDPC codes with parameters $[[N, \Theta(N), \Theta(N)]]$. These codes achieve linear distance $d = \Theta(N)$, which is impossible in 2D under Theorem 3.1 when the interaction range satisfies $L = O(1)$, the maximum stabilizer weight satisfies $w = O(1)$, and the rate satisfies $R = \Theta(1)$ (since then Theorem 3.1 implies $d = O(1)$). This is consistent with our results because these asymptotically good codes **cannot** be embedded in any fixed-dimensional lattice with local checks. Their interaction graphs have expansion properties that forbid low-dimensional embeddings (the geometric separator lemma fails). Our Theorem 3.1 essentially quantifies the "price of locality": to achieve linear distance, one must abandon fixed-dimensional geometric locality. At a high level, strong expansion is a key ingredient enabling linear-distance behavior in LDPC constructions; Theorem 3.3 records a standard expander-code implication on the classical side.

More precisely, geometric locality (bounded density and bounded interaction range) typically implies the existence of sublinear separators in the interaction graph (or related locality graphs): in two dimensions one expects separators of size $O(\sqrt{n})$ (up to model-dependent constants and coarse-graining at scale L), whereas constant-degree expanders have linear-size vertex separators. This separator-versus-expansion tension is one mechanism behind why expander-based LDPC constructions are incompatible with fixed-dimensional geometric locality; see, e.g., classical separator theorems [LT79, MTTV97].

5 Discussion and Open Problems

Our results advance the understanding of quantum LDPC code limitations under realistic hardware constraints, bridging the gap between abstract expander-based constructions and geometrically constrained implementations. However, several significant questions remain open.

1. **Lattice Geometry Dependence:** Can the dependence on the interaction range L and the constant C in Theorem 3.1 be improved for specific lattice geometries, such as hyperbolic tilings, which naturally support better expansion properties than Euclidean grids?
2. **Stoquastic Hamiltonians:** What is the optimal tradeoff when stoquasticity constraints are imposed on the Hamiltonian? Stoquastic Hamiltonians (those with non-positive off-diagonal matrix elements) are easier to simulate but may have more restrictive ground state properties relevant to code distance.
3. **Probabilistic Bounds:** How do these bounds change under probabilistic constructions? While worst-case bounds provide hard limits, random instantiations of local codes might exhibit better average-case behavior, potentially allowing for looser practical constraints.

6 Conclusion

In this work, we rephrased the 2D BPT locality tradeoff in a form that makes the dependence on interaction range and maximum stabilizer weight explicit, and we illustrated tightness via the toric code. We also highlighted why asymptotically good quantum LDPC codes must give up fixed-dimensional geometric locality, pointing to expansion as a recurring structural feature in such constructions.

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